

Lucas Narloch Zabla

Backend Developer | Python & FastAPI

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Backend Developer with production experience developing Python and FastAPI applications, RESTful APIs, internal tools, and automation solutions in AWS environments. Experienced in integrating backend services with relational databases, corporate systems, IoT devices, and external platforms. Focused on clean and maintainable code, automated testing, logging, troubleshooting, infrastructure automation, and continuous improvement.

TECHNICAL SKILLS

- **Backend:** Python, FastAPI, Node.js, TypeScript, NestJS, RESTful APIs, Object-Oriented Programming
- **Databases:** PostgreSQL, MySQL, DynamoDB
- **Testing & Quality:** pytest, Testcontainers, unit testing, integration testing, error handling, logging
- **Cloud & DevOps:** AWS Lambda, S3, SQS, EC2, RDS, IoT Core, CloudWatch, Terraform, GitHub Actions, Docker
- **Security:** JWT, OAuth 2.0, OpenID Connect, AWS Cognito
- **Tools:** Git, Linux, Bash, Nginx, Prometheus, Grafana

EXPERIENCE

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda. 09/2024 - Present

- Develop and maintain backend applications and RESTful APIs using Python and FastAPI in an AWS production environment.
- Developed an endpoint for data migration between IoT devices, integrating MySQL, DynamoDB, S3, and AWS IoT Core via MQTT, supporting the migration of thousands of historical records per operation.
- Implemented an automated test suite for FastAPI APIs with pytest, including 30 integration tests with Testcontainers and 80 unit tests, increasing delivery reliability and reducing the risk of production regressions.
- Investigated production issues through structured logs, metrics, and AWS CloudWatch, optimizing Python Lambda functions and reducing the average number of errors every 5 minutes from 113 to 5 (95.6%).
- Contributed to consolidating multiple serverless functions into a structured FastAPI application, reducing operational complexity, centralizing business rules, and improving system maintainability.
- Created automation solutions and asynchronous workflows with Amazon SQS, processing peaks above 40,000 notifications per day and eliminating blocking operations from the main request flow.
- Integrated backend services with relational databases, NoSQL databases, AWS services, IoT devices, and external authentication providers.
- Worked with JWT, OAuth 2.0, OpenID Connect, and AWS Cognito to implement authentication, authorization, and account integration flows.
- Automated deployments and environment configuration with GitHub Actions, Terraform, Bash scripts, and Docker, reducing manual steps and production configuration risks.
- Participated in technical discussions involving API design, system architecture, event-driven communication, observability, deployment, and production reliability.
- Collaborated with backend, frontend, firmware, and product teams to understand requirements and translate business needs into reliable technical solutions.

PROJECTS

Sunlog.dev

Production devlog platform with an architecture based on REST APIs, PostgreSQL, JWT authentication, asynchronous processing, and automated AWS infrastructure. Includes a backend with 33 HTTP endpoints, over 100 automated tests, CI/CD with GitHub Actions, Docker containerization, Nginx reverse proxy, structured logging, and observability with Prometheus and Grafana.

Technologies: NestJS, TypeScript, PostgreSQL, JWT, Docker, Nginx, AWS, GitHub Actions, Prometheus, Grafana, Next.js

[Code](#) | [Live demo](#)

PocScript

Programming language and compiler implemented in C, covering the full compilation pipeline: lexical analysis, AST parsing, semantic analysis, custom IR generation, LLVM IR emission, and native executable generation. Includes a custom runtime based on Linux syscalls, static type system, scope resolution, arrays, pointers, error handling, and a suite with 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux syscalls, Compiler Design

[Code](#)

EDUCATION

Bachelor of Computer Science - PUCPR

02/2025 - 12/2028

Full scholarship (100%). Coursework focused on algorithms, data structures, object-oriented programming, operating systems, computer networks, databases, and software engineering.

Technical High School in Electronics - CEEP

02/2022 - 12/2024

Technical education focused on electronics, embedded systems, digital systems, automation, and programming.

CERTIFICATIONS

- **Fundamentals of Database Engineering** - Hussein Nasser, Udemy | Sep/2025

LANGUAGES

- **Portuguese:** Native
- **English:** Professional Working Proficiency